

171. 800GMII Extender and 800GMII Extender Sublayer (800GXS)

171.1 Overview

This clause defines the functional characteristics for the optional 800GMII Extender and 800GMII Extender Sublayer (800GXS).

The 800GMII Extender allows the extension of the 800GMII to the PCS via a physical instantiation. The 800GMII Extender is composed of a DTE 800GXS at the RS end and a PHY 800GXS at the PHY end, with one or two 800GAUI-n between them.

Physical Layer clauses associated with the 800GMII Extender are listed in Table 171–1.

Table 171–1— Physical Layer clauses associated with the 800GMII Extender

Associated Clause	800GMII Extender
170—800GMII	Required
171—DTE 800GXS	Required
171—PHY 800GXS	Required
173—800GBASE-R PMA	Required
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a

^a An 800GMII Extender includes at least one 800GAUI-8.

Figure 171–1 shows the relationship of the 800GMII Extender and 800GXS with other sublayers to the ISO Open System Interconnection (OSI) reference model.

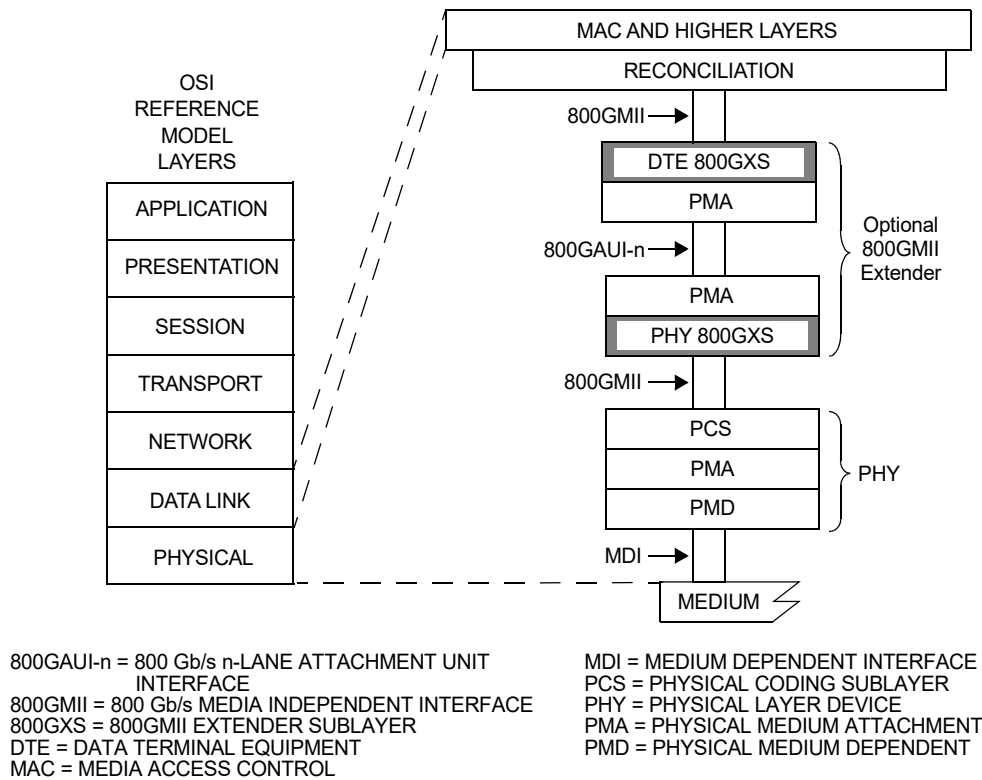


Figure 171-1—800GXS relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

171.1.1 Summary of functions

The following is a list of the major functions of the 800GMII Extender:

- Simple signal mapping to the 800GMII
- The optional 800GMII Extender can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 800GMII
- Independent transmit and receive data paths
- Each 800GXS has the same functions as the 800GBASE-R PCS (see Clause 172)
- Each 800GXS connects to an 800GAUI-n as shown in Figure 171-1

171.2 DTE 800GXS

The DTE 800GXS shall be identical in function to the 800GBASE-R PCS (see Clause 172) with the exception that the FEC degrade signaling is defined in 171.6. Figure 172-2 provides a functional block diagram.

171.2.1 DTE 800GXS inter-sublayer interfaces

The service interfaces used by the DTE 800GXS are identical to those defined in 172.1.5.

171.3 PHY 800GXS

The PHY 800GXS shall be identical in function to an 800GBASE-R PCS (see Clause 172) with the following exceptions:

- The PCS is inverted with the transmit function used for the receive direction and vice versa.
- The service interface signals are remapped as defined in 171.3.2 and 171.3.3.
- FEC degrade signaling is defined in 171.6.

Figure 171–2 provides a functional block diagram of the PHY 800GXS.

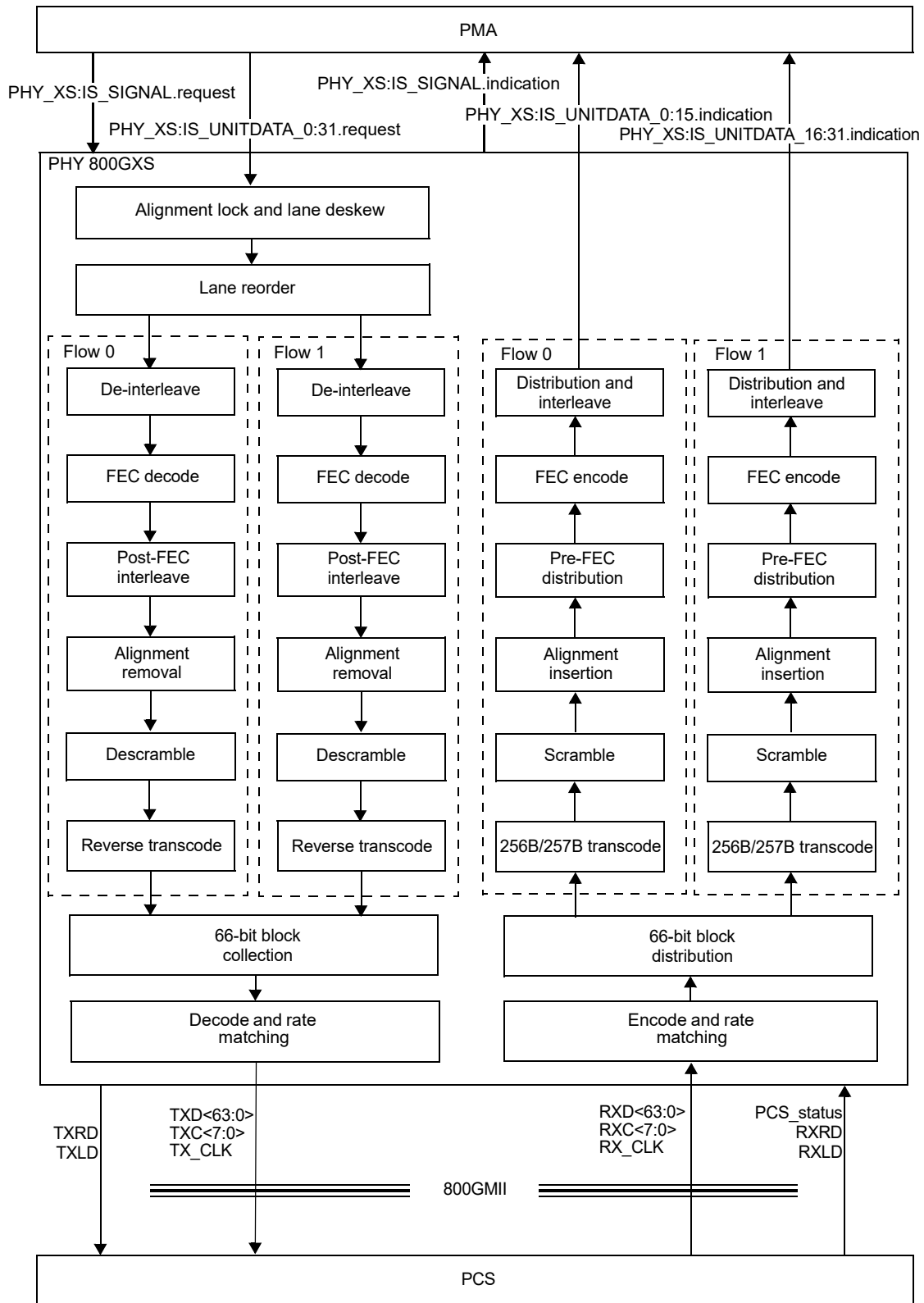


Figure 171–2—Functional block diagram for the PHY 800GXS

171.3.1 PHY 800GXS inter-sublayer interfaces

The PHY 800GXS inter-sublayer service interfaces are shown in Figure 169–3.

171.3.2 PHY 800GXS service interface

The PHY 800GXS service interface primitives are summarized as follows:

```
PHY_XS:IS_UNITDATA_i.request(tx_symbol)
PHY_XS:IS_UNITDATA_i.indication(rx_symbol)
PHY_XS:IS_SIGNAL.request(SIGNAL_OK)
PHY_XS:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to 31. In the transmit direction, the PHY 800GXS receives 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, from the PMA above. In the receive direction, the PHY 800GXS sends 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, to the PMA.

PHY_XS:IS_UNITDATA_i.request is the same as PMA:IS_UNITDATA_i.indication for the 32:8 PMA defined in 173.3.

PHY_XS:IS_UNITDATA_i.indication is the same as PMA:IS_UNITDATA_i.request for the 32:8 PMA defined in 173.3.

PHY_XS:IS_SIGNAL.request is the same as the PMA:IS_SIGNAL.indication for the 32:8 PMA defined in 173.3.

The PHY 800GXS provides signal status information to the PMA using the PHY_XS:IS_SIGNAL.indication(SIGNAL_OK) primitive (see Figure 171–2). The SIGNAL_OK parameter is set to OK when PCS_status (see 171.3.3) is true and is set to FAIL when PCS_status is false.

171.3.3 Service interface below the PHY 800GXS

The service interface below the PHY 800GXS allows the PHY 800GXS to transfer information to and from the PCS below.

The service interface below the PHY 800GXS is the 800GMII defined in Clause 170, with the following exceptions and additional signals:

- TX_CLK is used in place of RX_CLK, and vice versa
- TXD<63:0> is used in place of RXD<63:0>, and vice versa
- TXC<7:0> is used in place of RXC<7:0>, and vice versa
- Additional signals RXRD, RXLD, and PCS_status are provided in the receive direction (see 172.1.5.1)
- An additional signal TXRD indicates the state of the rx_rm_degraded variable (see 171.6.2) as detected by the PHY 800GXS in the transmit direction
- An additional signal TXLD indicates the state of the FEC_degraded_SER variable (see 171.6.2) as detected by the PHY 800GXS in the transmit direction

NOTE—In the transmit direction, if no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 800GMII will be reduced by approximately 49 ppm.

171.4 800GAUI-n

An 800GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers, as shown in Figure 171–1:

- 800GAUI-8 chip-to-chip (Annex 120F)
- 800GAUI-8 chip-to-module (Annex 120G)

171.5 Link fault signaling in the receive direction

Link fault signaling generated by the local PHY or by the link partner is sent toward the Reconciliation Sublayer as ordered sets through the 800GMII Extender.

The PHY 800GXS provides signal status from the local PHY to the client PMA using the PHY_XS:IS_SIGNAL.indication(SIGNAL_OK) primitive (see 171.3.2 and Figure 171–2).

171.6 FEC degrade

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in 116.6.

FEC degrade detection is specified in 172.2.5.3. FEC degrade detection is optional.

FEC degrade signaling using the PCS alignment marker status field is specified for the DTE 800GXS and PHY 800GXS in 171.6.1 and 171.6.2, respectively.

FEC degrade signaling across the service interface below the PHY 800GXS is specified in 171.3.3.

171.6.1 DTE 800GXS FEC degrade signaling

The variable `tx_am_sf` is set as follows:

`tx_am_sf<2:0> = {FEC_degraded_SER + rx_local_degraded, 0, 0}`
where `FEC_degraded_SER` and `rx_local_degraded` are defined in 172.2.6.2.2 and + denotes logical OR.

171.6.2 PHY 800GXS FEC degrade signaling

The variable `tx_am_sf` is set as follows:

`tx_am_sf<2:0> = {RXRD, RXLD, 0}`
where `RXRD` and `RXLD` are signals from the PCS on the PCS service interface (see 172.1.5.1).

171.7 800GXS partitioning example

Two partitioning examples and MMD numbering using the 800GXS are shown in Figure 171–3. Annex 173A shows additional examples of 800GXS partitioning and MMD numbering.

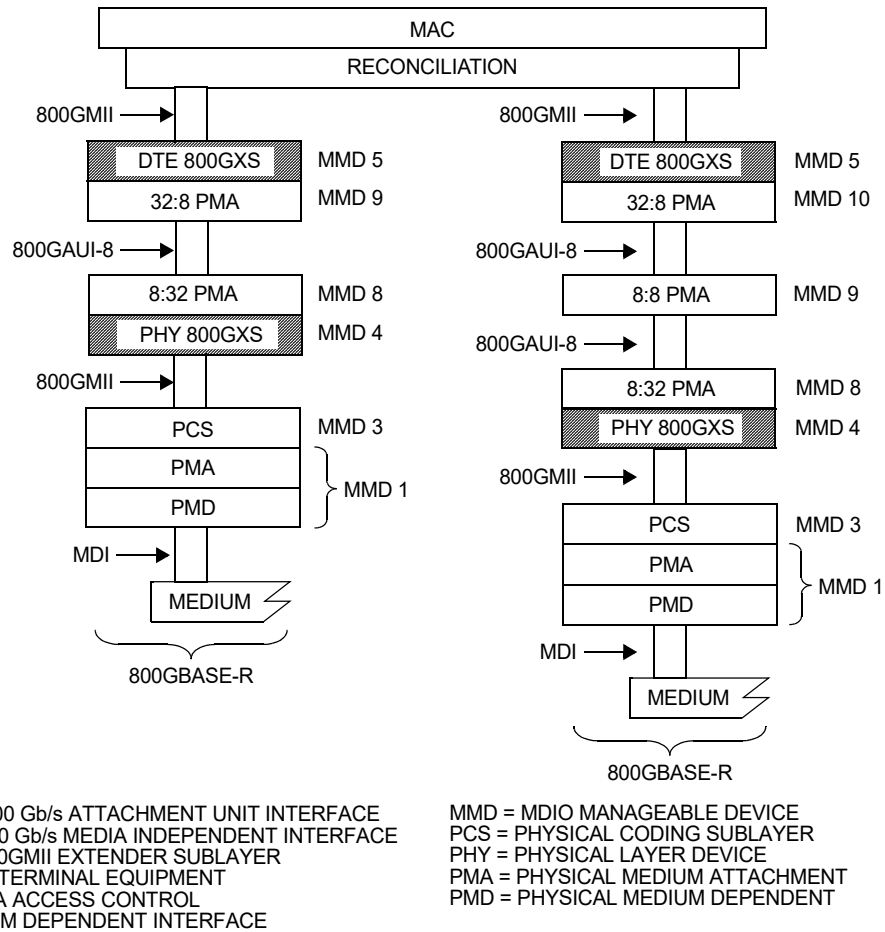


Figure 171-3—Example 800GBASE-R PMA layering with 800GXS

171.8 800GXS MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the 800GXS. If MDIO is implemented, it shall map MDIO PHY 800GXS and DTE 800GXS control bits to Clause 172 control variables as shown in Table 171-2 and Table 171-4, respectively. Similarly, if MDIO is implemented, it shall map MDIO PHY 800GXS and DTE 800GXS status bits to Clause 172 status variables as shown in Table 171-3 and Table 171-5, respectively. MDIO registers relevant to the PHY 800GXS and DTE 800GXS are described in 45.2.4 and 45.2.5, respectively.

Table 171–2—MDIO PHY 800GXS to Clause 172 control variable mapping

MDIO control variable	PHY XS register name	Register/bit number	Clause 172 control variable
Loopback	PHY XS control 1 register	4.0.14	Loopback
Reset	PHY XS control 1 register	4.0.15	reset
Transmit test-pattern enable	BASE-R PHY XS test-pattern control register	4.42.3	tx_test_mode
PHY XS FEC bypass indication enable	PHY XS FEC control register	4.800.1	FEC_bypass_indication_enable
PHY XS FEC degraded SER enable	PHY XS FEC control register	4.800.2	FEC_degraded_SER_enable
PHY XS FEC degraded SER activate threshold	PHY XS FEC degraded SER activate threshold register	4.806, 4.807	FEC_degraded_SER_activate_threshold
PHY XS FEC degraded SER deactivate threshold	PHY XS FEC degraded SER deactivate threshold register	4.808, 4.809	FEC_degraded_SER_deactivate_threshold
PHY XS FEC degraded SER interval	PHY XS FEC degraded SER interval	4.810, 4.811	FEC_degraded_SER_interval

Table 171–3—MDIO PHY 800GXS to Clause 172 status variable mapping

MDIO status variable	PHY XS register name	Register/bit number	Clause 172 status variable
BASE-R PHY XS receive link status	BASE-R PHY XS status 1	4.32.12	PCS_status
PHY XS lane alignment status	Multi-lane BASE-R PHY XS alignment status 1	4.50.12	align_status
Lane 0 to 31 aligned	Multi-lane BASE-R PHY XS alignment status 3, 4, and 5	4.52.7:0 4.53.15:0 4.54.7:0	amps_lock<0:31>
FEC codeword counter	PHY XS RS-FEC codeword counter	4.300 to 4.302	FEC_cw_counter
FEC codeword error bin counter, 1 to 15	PHY XS RS-FEC codeword error bin, 1 to 15	4.340 to 4.369	FEC_codeword_error_bin<1:15>
Lane 0 to 31 mapping	PHY XS lane mapping, lane 0 through lane 31	4.400 through 4.431	pcs_lane_mapping<0:31>
PHY XS FEC symbol errors, lane 0 to lane 31	PHY XS FEC symbol error counter, lane 0 to lane 31	4.600 to 4.663	FEC_symbol_error_counter<0:31>
PHY XS FEC bypass indication ability	PHY XS FEC status	4.801.1	FEC_bypass_indication_ability
PHY XS FEC high SER	PHY XS FEC status register	4.801.2	hi_ser_combined
PHY XS FEC degraded SER ability	PHY XS FEC status register	4.801.3	FEC_degraded_SER_ability

Table 171–3—MDIO PHY 800GXS to Clause 172 status variable mapping (*continued*)

MDIO status variable	PHY XS register name	Register/bit number	Clause 172 status variable
PHY XS FEC degraded SER	PHY XS FEC status register	4.801.4	FEC_degraded_SER
Remote degraded SER received	PHY XS FEC status register	4.801.5	rx_rm_degraded
Local degraded SER received	PHY XS FEC status register	4.801.6	rx_local_degraded
FEC corrected codewords	PHY XS FEC corrected codewords counter	4.802, 4.803	FEC_corrected_cw_counter
FEC uncorrected codewords	PHY XS FEC uncorrected codewords counter	4.804, 4.805	FEC_uncorrected_cw_counter

Table 171–4—MDIO DTE 800GXS to Clause 172 control variable mapping

MDIO control variable	DTE XS register name	Register/bit number	Clause 172 control variable
Loopback	DTE XS control 1 register	5.0.14	Loopback
Reset	DTE XS control 1 register	5.0.15	reset
Transmit test-pattern enable	BASE-R DTE XS test-pattern control register	5.42.3	tx_test_mode
DTE XS FEC bypass indication enable	DTE XS FEC control register	5.800.1	FEC_bypass_indication_enable
DTE XS FEC degraded SER enable	DTE XS FEC control register	5.800.2	FEC_degraded_SER_enable
DTE XS FEC degraded SER activate threshold	DTE XS FEC degraded SER activate threshold register	5.806, 5.807	FEC_degraded_SER_activate_threshold
DTE XS FEC degraded SER deactivate threshold	DTE XS FEC degraded SER deactivate threshold register	5.808, 5.809	FEC_degraded_SER_deactivate_threshold
DTE XS FEC degraded SER interval	DTE XS FEC degraded SER interval	5.810, 5.811	FEC_degraded_SER_interval

Table 171–5—MDIO DTE 800GXS to Clause 172 status variable mapping

MDIO status variable	DTE XS register name	Register/bit number	Clause 172 status variable
BASE-R DTE XS receive link status	BASE-R DTE XS status 1	5.32.12	PCS_status
DTE XS lane alignment status	Multi-lane BASE-R DTE XS alignment status 1	5.50.12	align_status
Lane 0 to 31 aligned	Multi-lane BASE-R DTE XS alignment status 3,4, and 5	5.52.7:0 5.53.15:0 5.54.7:0	amps_lock<0:31>
FEC codeword counter	DTE XS RS-FEC codeword counter	5.300 to 5.302	FEC_cw_counter
FEC codeword error bin counter, 1 to 15	DTE XS RS-FEC codeword error bin, 1 to 15	5.340 to 5.369	FEC_codeword_error_bin_<1:15>
Lane 0 to 31 mapping	DTE XS lane mapping, lane 0 through lane 31	5.400 through 5.431	pcs_lane_mapping<0:31>
DTE XS FEC symbol errors, lane 0 to lane 31	DTE XS FEC symbol error counter, lane 0 to lane 31	5.600 to 5.663	FEC_symbol_error_counter<0:31>
DTE XS FEC bypass indication ability	DTE XS FEC status	5.801.1	FEC_bypass_indication_ability
DTE XS FEC high SER	DTE XS FEC status register	5.801.2	hi_ser_combined
DTE XS FEC degraded SER ability	DTE XS FEC status register	5.801.3	FEC_degraded_SER_ability
DTE XS FEC degraded SER	DTE XS FEC status register	5.801.4	FEC_degraded_SER
Remote degraded received	DTE XS FEC status register	5.801.5	rx_rm_degraded
Local degraded received	DTE XS FEC status register	5.801.6	rx_local_degraded
FEC corrected codewords	DTE XS FEC corrected codewords counter	5.802, 5.803	FEC_corrected_cw_counter
FEC uncorrected codewords	DTE XS FEC uncorrected codewords counter	5.804, 5.805	FEC_uncorrected_cw_counter

171.9 Protocol implementation conformance statement (PICS) proforma for Clause 171, 800GMII Extender and 800GMII Extender Sublayer (800GXS)¹²

171.9.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 171, 800GMII Extender, and 800GMII Extender Sublayer (800GXS), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

171.9.2 Identification

171.9.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

171.9.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 171, 800GMII Extender and 800GMII Extender Sublayer (800GXS)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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¹²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

171.9.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
MII	800GMII logical interface is supported	172.1.5.1		O	Yes [] No []
*DTEXS	DTE 800GXS	171.2		O/2	Yes [] No []
*PHYXS	PHY 800GXS	171.3		O/2	Yes [] No []
*FDD	Support for FEC degrade detection	171.6		O	Yes [] No []
*MD	MDIO	171.8		O	Yes [] No []

171.9.4 PICS proforma tables for 800GMII Extender and 800GMII Extender Sublayer (800GXS)

171.9.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	172.2.4.1.2		O	Yes [] No []
TF1	Transmit 64B/66B encoder uses state diagram in Figure 119–14	172.2.4.1		!SE:M	Yes [] N/A []
TF2	Transmit 64B/66B encoder uses stateless rules in Table 172–1	172.2.4.1.2		SE:M	Yes [] N/A []
TF3	64B/66B to 256B/257B transcoder	172.2.4.4		M	Yes []
TF4	Scrambler	172.2.4.5		M	Yes []
TF5	Pad value	172.2.4.6		M	Yes []
TF6	Alignment marker insertion	172.2.4.6		M	Yes []
TF7	FEC degrade signaling	172.2.4.6		M	Yes []
TF8	Pre-FEC distribution	172.2.4.7		M	Yes []
TF9	Reed-Solomon encoder	172.2.4.8		M	Yes []
TF10	Symbol distribution	172.2.4.9		M	Yes []
TF11	Transmission bit ordering	172.2.4.10		M	Yes []
TF12	Scrambled idle transmit test-pattern generator	172.2.4.11		M	Yes []

171.9.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
*BI	Bypass indication supported	172.2.5.3		O	Yes [] No []
*SD	Uses stateless decoder	172.2.5.9.2		O	Yes [] No []
RF1	Alignment marker lock	172.2.5.1		M	Yes []
RF2	PCS lane number is captured to an MDIO register	119.2.6.3		MD:M	Yes [] N/A []
RF3	Deskew	172.2.5.1		M	Yes []
RF4	Lane reorder and de-interleave	172.2.5.2		M	Yes []
RF5	Reed-Solomon decoder corrects any combination of up to $t=15$ symbol errors in a codeword	172.2.5.3		M	Yes []
RF6	Reed-Solomon decoder is capable of indicating when a codeword was not corrected	172.2.5.3		M	Yes []
RF8	Error monitoring and assertion of hi_ser, while error indication is bypassed	172.2.5.3		BI:M	Yes [] N/A []
RF9	Detect FEC degrade	172.2.5.3		FDD:M	Yes [] N/A []
RF10	Alignment marker removal	172.2.5.5		M	Yes []
RF11	Descrambler	172.2.5.6		M	Yes []
RF12	256B/257B to 64B/66B transcoder	172.2.5.7		M	Yes []
RF13	Receive 64B/66B decoder uses state diagram in Figure 119–15	172.2.5.9.1		!SD:M	Yes [] N/A []
RF14	Receive 64B/66B decoder uses stateless rules in Table 172–4	172.2.5.9.2		SD:M	Yes [] N/A []

171.9.4.3 64B/66B coding rules

Item	Feature	Subclause	Value/Comment	Status	Support
C1	Encoder (and ENCODE function) implements the code as specified	172.2.3, 172.2.6.2.3		M	Yes []
C2	Decoder (and DECODE function) implements the code as specified	172.2.3, 172.2.6.2.3		M	Yes []
C3	Only valid block types are transmitted	172.2.3		M	Yes []
C4	Invalid block types are treated as an error	172.2.3		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
C5	Only valid control characters are transmitted	172.2.3		M	Yes []
C6	Invalid control characters are treated as an error	172.2.3		M	Yes []
C7	Idles do not interrupt data	172.2.3		M	Yes []
C8	IDLE control code insertion and deletion	172.2.3	Insertion or deletion in groups of 8 /I/s	M	Yes []
C9	Sequence ordered set deletion	172.2.3	Only one whole ordered set of two consecutive sequence ordered sets may be deleted	M	Yes []

171.9.4.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to XS management objects is provided	172.3		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits for PHY 800GXS	171.8	See Table 118–1 and Table 118–2	MD*PHYXS:M	Yes [] N/A []
M3	Mapping of MDIO control bits and MDIO status bits for DTE 800GXS	171.8	See Table 118–3 and Table 118–4	MD*DTEXS:M	Yes [] N/A []

171.9.4.5 Loopback

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Supports loopback	172.4		M	Yes []
L2	When in loopback, transmits what it receives from the 800GMII	172.4		M	Yes []
L3	When in loopback, ignore all data presented by the PMA	172.4		M	Yes []

171.9.4.6 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1	800GXS delay constraint	172.5		M	Yes []